

Unit 4, Lesson 9: Algebraic Expressions with Negative Exponents

Benchmark

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Learning Target

☐ I can write an equivalent algebraic expression by applying the laws of exponents.

4.9.1 Warm-Up: Bell Work

1. For each expression, apply the laws of exponents to generate an equivalent expression with the fewest factors possible.

a. $\frac{k^{11}}{k^6} = \underline{\hspace{2cm}}$

b. $\frac{-2g^5k^6}{10gk^3} = \underline{\hspace{2cm}}$

c. $x^3 \cdot x^8 = \underline{\hspace{2cm}}$

d. $\left(\frac{1}{3}x^2 \cdot y\right)^4 = \underline{\hspace{2cm}}$

4.9.2 Collaboration: Negative Exponents in Algebraic Expressions

1. Rewrite each expression using positive exponents.

$$6^{-4} = \underline{\hspace{2cm}}$$

$$(-3)^{-2} = \underline{\hspace{2cm}}$$

$$\left(\frac{4}{5}\right)^{-6} = \underline{\hspace{2cm}}$$

$$\left(\frac{-7}{8}\right)^{-1} = \underline{\hspace{2cm}}$$

2. Make a conjecture on how to rewrite an algebraic expression with a negative exponent.

$$x^{-3} = \underline{\hspace{2cm}}$$

3. Work with your partner to complete the following to explore negative exponents in algebraic expressions.

- a. Rewrite each expression in expanded form.

Algebraic Expression	Expanded Form
x^3	
x^2	
x^1	
x^0	

$\curvearrowright$
 $\curvearrowright$
 $\curvearrowright$

$\times \frac{\boxed{}}{\boxed{}}$
 $\times \frac{\boxed{}}{\boxed{}}$
 $\times \frac{\boxed{}}{\boxed{}}$

- b. Discuss with your partner what factor could be multiplied by each expression to generate the expression in the next row. Then, fill in the blanks to the right of the table.

- c. Complete the statement.

As the value of the exponent of x
☐ increases
☐ decreases
 by 1 in the column on the left, the equivalent expression in the column on the right can be found by multiplying the previous value by the
☐ opposite
☐ reciprocal
 of x .

- d. Continue the pattern.

Algebraic Expression	Expanded Form	Equivalent Expression
x^0	$\frac{x}{x}$	1
x^{-1}		
x^{-2}		
x^{-3}		



$\cdot \frac{1}{x}$

$\cdot \frac{1}{x}$

$\cdot \frac{1}{x}$

- e. Compare your conjecture in question 2 with the expression you found in Part D.
- f. Complete the statement.

A variable base with a negative exponent, $(x)^{-n}$, can be rewritten using the
☐ reciprocal
☐ opposite
 of the base with the
☐ reciprocal
☐ opposite
 exponent.

4. For each expression, write an equivalent expression with positive exponents.

a. $x^{-4} = \underline{\hspace{2cm}}$

b. $c^{-9} = \underline{\hspace{2cm}}$

c. $\left(\frac{1}{f}\right)^{-2} = \underline{\hspace{2cm}}$

d. $\left(\frac{a}{a}\right)^{-5} = \underline{\hspace{2cm}}$

4.9.3 Activity: Equivalent Expressions Card Sort

1. Work with your partner or group to match the equivalent expressions in the set of cards. Each numbered card in the set shown below has three equivalent matches. Record the letters of the matching cards in the table.

Numbered Cards	Equivalent Expression Cards
1 x^7	
2 x^{12}	
3 x^{10}	
4 x^{-9}	
5 $\frac{1}{x^6}$	
6 $\frac{1}{x^{11}}$	
7 x^4	
8 x^{-5}	
9 x^8	

4.9.4 Your Turn!

1. For each expression, generate an equivalent expression with positive exponents by applying one or more law of exponents.

a. $7x^5 =$ _____

b. $3x^{-4} - 2x^{-6} =$ _____

2. Find the value of m that makes the equation true.

$$\frac{x^4}{x^m} = x^{10}$$

4.9.5 Wrap-Up: Reflection

1. Draw an **X** on the line to indicate how you feel about the learning target.

☐ I can write an equivalent algebraic expression by applying the laws of exponents.

I need help.

I'm getting there, but

I understand this

I can't get started.

I need more practice.

and I feel confident.

